

Conductance and Kernel Choices for Graph Laplacians: Theory, Tasks, and Implications for Graph Low-Pass Smoothing

SIMODS Provenance/Librarian synthesis from H005 reviewer-auditor memos

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Abstract

This report synthesizes ten full-paper reviews and audits on graph conductance and kernel choices for Laplacians, diffusion operators, semi-supervised graph regression, manifold learning, and graph signal smoothing. The immediate SIMODS purpose is methodological: to guide a future benchmark that compares the current `gflow` low-pass/GCV smoother with length- and kernel-conductance alternatives. The central conclusion is that researchers rarely choose conductances only as inverse metric lengths. Gaussian kernels, heat kernels, self-tuned/local-scale kernels, density-normalized kernels, Markov normalizations, and graph-spectral filters each encode a different answer to the question “what should count as a smooth signal on this graph?” These choices change the graph energy, the eigenbasis, the diffusion geometry, and therefore the behavior of low-pass smoothing and GCV. For SIMODS, the current `fit.rdgraph.regression()` operator must be kept conceptually separate from these planned comparators: the current precomputed-graph path uses supplied edge lengths for local neighborhood ordering/truncation and then constructs an overlap-density/Riemannian-complex conductance, not direct inverse-length, Gaussian, or self-tuned conductances.

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1 Review Corpus and Evidence Conventions

This synthesis is based on the H005 review workspace `literature/conductance_kernel_laplacian_review`. The source manifest records the canonical reading copy for each Tier 1 paper, including original URL, DOI or arXiv identifier when available, local path, access date, page count, and SHA-256. Each Tier 1 paper was assigned to a dedicated reviewer and a dedicated auditor. Reviewers read the whole paper, including figures. Auditors checked formulas, operator definitions, figure references, evidence labels, and relevance to `gflow`/SIMODS. All ten audits returned minor revisions; no auditor requested a full reread. The review memos now include post-audit revision notes.

The evidence labels used here are:

explicit The claim is directly stated in the reviewed paper.

derived The claim follows from formulas, algorithms, figures, or experiments in the paper, but is not stated in exactly this language.

contextual The claim is our SIMODS/`gflow` interpretation or cross-paper synthesis.

uncertain The claim is plausible but ambiguous, typographically suspect, or awaiting developer/paper confirmation.

Unless otherwise marked, paper-specific statements in the paper-by-paper section are summaries of explicit or directly derived review findings. The SIMODS recommendations are contextual.

Table 1: Tier 1 corpus and review role in the synthesis.

ID	Paper	Main conductance/kernel theme	Main SIMODS role
P01	Zelnik-Manor and Perona [2004]	self-tuned local-scale Gaussian affinity	adaptive/local-scale comparator
P02	Belkin and Niyogi [2001]	heat-kernel and binary graph weights for Laplacian Eigenmaps	heat-kernel manifold-learning baseline
P03	Coifman and Lafon [2006]	kernel, density normalization, Markov diffusion	diffusion/Markov normalization baseline
P04	Berry and Harlim [2016]	variable-bandwidth diffusion kernels	local-bandwidth and density-correction theory
P05	Zhu et al. [2003]	graph weights for harmonic-function semi-supervised learning	graph interpolation/regression analogue
P06	Belkin et al. [2006]	graph Laplacian regularization in learning	regularization objective analogue

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ID	Paper	Main conductance/kernel theme	Main SIMODS role
P07	Shuman et al. [2013]	graph Fourier bases and spectral filters	graph low-pass smoothing semantics
P08	Hein et al. [2007]	convergence of graph Laplacians on random neighborhood graphs	normalization and density-limit warnings
P09	Belkin and Niyogi [2008]	theoretical foundation for Laplacian manifold methods	heat-kernel/Laplacian Beltrami theory baseline
P10	Cheng and Wu [2022]	kNN self-tuned kernel graph Laplacian convergence	adaptive/kNN local-scale theory

2 Primer: From Lengths to Conductances to Smoothers

2.1 Weighted Graph Laplacians

Let $G = (V, E)$ be an undirected graph with $n = |V|$ vertices. A graph signal is a vector $f \in \mathbb{R}^n$, or a matrix $F \in \mathbb{R}^{n \times p}$ when p features are smoothed on the same graph. A nonnegative edge weight

$$c_{ij} = c_{ji} \geq 0$$

can be called an affinity, similarity, weight, or conductance depending on the paper and task. With weighted adjacency $W = (c_{ij})$, degree $D_{ii} = \sum_j c_{ij}$, and unnormalized Laplacian $L = D - W$, the standard Dirichlet energy is

$$\mathcal{E}_G(f) = f^\top L f = \frac{1}{2} \sum_{i,j} c_{ij} (f_i - f_j)^2 = \sum_{\{i,j\} \in E} c_{ij} (f_i - f_j)^2.$$

The factor convention matters. Several reviewed papers print unhalved double sums while also writing $f^\top L f$. This does not change minimizers, but it does matter for precise comparisons of objectives and constants.

For a matrix signal F , the analogous energy is

$$\text{tr}(F^\top L F) = \frac{1}{2} \sum_{i,j} c_{ij} \|F_i - F_j\|^2.$$

Large c_{ij} makes disagreement across edge (i, j) expensive. Thus the conductance rule determines what functions are low energy before any smoothing parameter is chosen.

2.2 A Common Family of Conductance Rules

The review treats many formulas as members of a common family

$$c_{ij} = \phi(\ell_{ij}; \theta),$$

where ℓ_{ij} is an ambient distance, graph edge length, or local dissimilarity, and θ includes bandwidth, exponent, local bandwidth, density, or normalization parameters. Important examples are:

$$c_{ij}^{\text{len}} = (\ell_{ij} + \epsilon)^{-\alpha}, \quad (1)$$

$$c_{ij}^{\text{exp}} = \exp(-\ell_{ij}/\sigma), \quad (2)$$

$$c_{ij}^{\text{rbf}} = \exp(-\ell_{ij}^2/\sigma^2), \quad (3)$$

$$c_{ij}^{\text{self}} = \exp\left(-\frac{\ell_{ij}^2}{\sigma_i \sigma_j}\right), \quad (4)$$

$$c_{ij}^{\text{vb}} = h\left(\frac{\ell_{ij}^2}{\epsilon \rho_i \rho_j}\right). \quad (5)$$

The inverse-length rule is natural when edge length is interpreted as physical resistance or cost. Heat/RBF kernels are natural when the graph is meant to approximate local heat diffusion on a point cloud or manifold. Self-tuned and variable-bandwidth kernels are natural when a single global scale fails under nonuniform sampling. Density-normalized kernels are natural when the target is geometry rather than sampling density.

2.3 Low-Pass Filtering

If $L = V\Lambda V^\top$ is a symmetric graph Laplacian eigendecomposition, a spectral graph filter has the form

$$S_h = Vh(\Lambda)V^\top,$$

where $h(\lambda)$ attenuates high-frequency modes. Heat smoothing uses $h_t(\lambda) = \exp(-t\lambda)$. Tikhonov smoothing uses

$$h_\eta(\lambda) = \frac{1}{1 + \eta\lambda}$$

or an equivalent parameterization. More general graph signal processing filters may be polynomial, idealized low-pass, Butterworth-like, or learned, but they all inherit their meaning from the graph eigenbasis.

If the operator is a row-stochastic diffusion matrix P with eigenvalues λ_j , diffusion powers use $h_t(\lambda_j) = \lambda_j^t$. This is low-pass relative to the Markov operator: slowly varying modes have eigenvalues near 1 and survive many diffusion steps.

2.4 Current gflow Smoother Versus Planned Comparators

The H003/H004/H005 line of work needs one hard distinction. The current `fit.rdgraph.regression()` precomputed-graph path does not convert supplied edge lengths directly into conductances by

$$c_e = \frac{1}{\ell_e + \epsilon}.$$

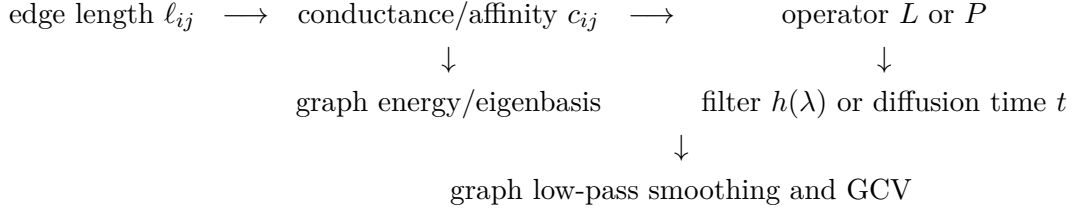
As established in the companion source audit, supplied `weight.list` values are positive graph edge lengths used for local neighborhood ordering and truncation. The mass-symmetrized spectral smoother then uses an overlap-density/Riemannian-complex edge mass $\rho_1(e)$, with conductance

$$c_e^\rho = \frac{1}{\max(\rho_1(e), 10^{-10})}.$$

Therefore the papers reviewed here mainly support planned comparator families: inverse-length, heat/RBF, exponential, self-tuned, variable-bandwidth, density-normalized, and Markov diffusion smoothers. They do not retroactively describe the current `gflow` smoother.

3 Concept Map

The concepts in this review form a pipeline, not a list of interchangeable phrases:



The main branches are:

- **Inverse length conductance** treats short edges as strong couplings and long edges as weak couplings. It is close to physical resistance/conductance intuition and directly tests graph metric geometry.
- **Heat kernels** convert squared local distances into local diffusion affinities. They are central in Laplacian Eigenmaps and Laplace-Beltrami approximation.
- **Self-tuned/local-scale kernels** replace a global bandwidth with endpoint scales such as $\sigma_i\sigma_j$, reducing global-scale fragility under nonuniform sampling.
- **Adaptive-radius/cKNN graph construction** changes the support of the graph. This is related to, but not identical to, self-tuned conductance weighting. A paper may use nearest-neighbor distances for bandwidths without defining cKNN support.
- **Diffusion/Markov normalization** changes affinities into a random-walk operator P . The resulting eigenfunctions and smoothing behavior depend on degree and density normalization.
- **Graph low-pass smoothing** applies a spectral response $h(\lambda)$ to the operator eigensystem. P07 is the clearest source for this filter semantics.

4 Paper-by-Paper Reviews

4.1 P01: Self-Tuning Spectral Clustering

Zelnik-Manor and Perona [2004] address a concrete failure mode of spectral clustering: one global Gaussian bandwidth can be too small for sparse regions and too large for dense regions. The paper modifies the Ng-Jordan-Weiss pipeline by replacing the global Gaussian affinity

$$A_{ij} = \exp\left(-\frac{d^2(s_i, s_j)}{\sigma^2}\right)$$

with the self-tuned affinity

$$\hat{A}_{ij} = \exp\left(-\frac{d^2(s_i, s_j)}{\sigma_i\sigma_j}\right), \quad \sigma_i = d(s_i, s_K),$$

where s_K is the K th neighbor of s_i . The normalized affinity operator is $L = D^{-1/2}\hat{A}D^{-1/2}$. In the paper's terminology, $\hat{A}$ is an affinity; interpreting it as a conductance is our graph smoothing translation.

The paper’s figures make the scale argument vivid. Figure 1 shows global bandwidth sensitivity. Figure 2 shows the central local-scale effect: local scaling weakens cross-cluster affinities that are misleading under a single global bandwidth. Figures 3–6 cover synthetic clustering, eigenvalue-gap limitations, alignment costs for estimating the number of clusters, and image segmentation.

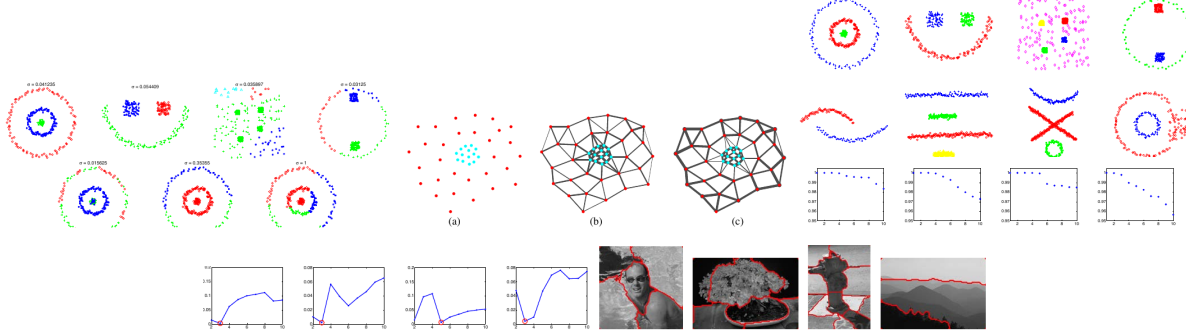


Figure 1: Cropped figure panels containing the P01 figures discussed above. Reproduced from Zelnik-Manor and Perona (2004), Figures 1–6, PDF pp. 2–7, from the canonical reading copy, for internal review only.

For SIMODS, P01 is the prototype for a self-tuned Gaussian comparator. It does not define cKNN support or adaptive-radius graph construction. Its nearest-neighbor distances set bandwidths, not necessarily graph support. Thus a SIMODS cKNN connection is contextual synthesis. The reviewer/auditor also flagged an ambiguity: Eq. (3) as printed gives an ideal per-row cost of 1, whereas Figure 5 plots near-zero alignment costs. The final manuscript should not rely on the exact plotted cost scale.

4.2 P02: Laplacian Eigenmaps

Belkin and Niyogi [2001] give one of the clearest early pipelines from point-cloud neighborhoods to graph Laplacian eigenvectors. Build a graph by an ϵ -neighborhood rule or symmetrized nearest-neighbor rule; assign either heat weights

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{t}\right)$$

or binary weights; solve

$$Ly = \lambda Dy, \quad L = D - W;$$

and use the first nontrivial eigenvectors as embedding coordinates.

The paper’s heat-kernel derivation prints the short-time local heat kernel

$$H_t(x, y) \approx (4\pi t)^{-n/2} \exp\left(-\frac{\|x - y\|^2}{4t}\right).$$

Thus the paper contains both t and $4t$ bandwidth conventions. They should be preserved as printed rather than silently merged.

The core variational identity is

$$y^\top Ly = \frac{1}{2} \sum_{i,j} (y_i - y_j)^2 W_{ij}.$$

For vector-valued embeddings,

$$\text{tr}(Y^\top LY) = \frac{1}{2} \sum_{i,j} \|Y_i - Y_j\|^2 W_{ij}.$$

The paper sometimes writes unhalved objectives; this is an irrelevant constant for minimizers but should be stated. P02 is a strong source for heat weights and graph smoothness, but not for density correction, local bandwidth, or current `gflow` overlap-density smoothing.

4.3 P03: Diffusion Maps

Coifman and Lafon [2006] move from symmetric affinities to Markov diffusion geometry. Starting with a symmetric nonnegative kernel $k(x, y)$, define

$$d(x) = \int k(x, y) d\mu(y), \quad p(x, y) = \frac{k(x, y)}{d(x)}.$$

The row-stochastic operator P has powers P^t , and the diffusion distance is the distance between t -step transition distributions. If $P\psi_\ell = \lambda_\ell \psi_\ell$, then

$$D_t(x, y)^2 = \sum_{\ell \geq 1} \lambda_\ell^{2t} (\psi_\ell(x) - \psi_\ell(y))^2,$$

and the diffusion map uses coordinates

$$\Psi_t(x) = (\lambda_1^t \psi_1(x), \dots, \lambda_s^t \psi_s(x)).$$

This paper is crucial because it separates three issues: the kernel, the degree/Markov normalization, and the diffusion time. Section 3 introduces α -normalization,

$$k_\epsilon^{(\alpha)}(x, y) = \frac{k_\epsilon(x, y)}{q_\epsilon(x)^\alpha q_\epsilon(y)^\alpha},$$

which changes the limiting operator. In particular, $\alpha = 1$ removes sampling-density bias in the Laplace-Beltrami limit, while other values retain density-dependent terms.

For graph low-pass smoothing, P03 gives the Markov counterpart of spectral filtering: P^t multiplies mode ℓ by λ_ℓ^t , damping small-eigenvalue modes of P . Calling Figure 3 a graph-signal low-pass figure is derived synthesis, not the paper’s own terminology. The audit also flagged the Section 3.4 $\alpha = 0$ phrase as an apparent typo because the heading and proposition concern $\alpha = 1$.

4.4 P04: Variable Bandwidth Diffusion Kernels

Berry and Harlim [2016] ask what changes when the diffusion kernel bandwidth is local:

$$K_\epsilon^S(x, y) = h\left(\frac{\|x - y\|^2}{\epsilon \rho(x) \rho(y)}\right).$$

The paper’s practical Gaussian version uses

$$K_\epsilon^S(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{4\epsilon \rho(x_i) \rho(x_j)}\right).$$

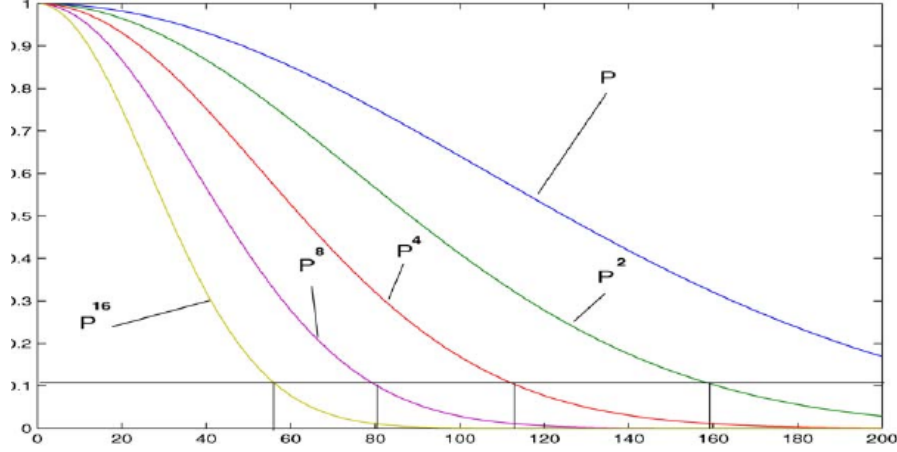


Figure 2: Cropped figure panel containing the P03 Figure 3 spectrum plot discussed as derived low-pass evidence. Reproduced from Coifman and Lafon (2006), Figure 3, PDF p. 12, from the canonical reading copy, for internal review only.

The bandwidth ρ is often taken as q^β , where q is sampling density. The limiting operator has the form

$$L_{\alpha,\rho}f = \Delta f + 2(1 - \alpha) \frac{\nabla f \cdot \nabla q}{q} + (d + 2) \frac{\nabla f \cdot \nabla \rho}{\rho}.$$

When $\rho = q^\beta$, this becomes

$$L_{\alpha,\beta}f = \Delta f + c_1 \frac{\nabla f \cdot \nabla q}{q}, \quad c_1 = 2 - 2\alpha + d\beta + 2\beta.$$

The audit notes that sign conventions for Δ must be stated when comparing this formula to graph smoothing or heat semigroups.

P04 is important for SIMODS because it shows that local scale is not just a numerical patch. It changes the continuum operator unless the normalization parameters are chosen deliberately. It supports variable-bandwidth kernel-conductance comparators and density-correction diagnostics. It does not describe the current `fit.rdgraph.regression()` overlap-density operator.

4.5 P05: Gaussian Fields and Harmonic Functions

Zhu et al. [2003] treat graph weights as the basis for semi-supervised interpolation. Given labeled and unlabeled vertices, the harmonic solution minimizes graph energy subject to labeled boundary values, leading to a Laplacian block system. The weights are similarities/affinities, and in the electrical-network interpretation they may be read as conductances. Outside that interpretation, the safer phrase is “weight/affinity, interpretable as conductance.”

The paper connects harmonic functions, Gaussian random fields, graph cuts, and kernel classifiers. The relevant SIMODS lesson is direct: changing the edge weights changes the interpolation. If labels or feature values are graph signals, the graph weights define what it means for the prediction to vary smoothly away from observed values.

The audit corrected two convention points. First, the heat-kernel sign should be written as $K_t = e^{-t\Delta}$ under the paper’s convention. Second, the Rayleigh quotient numerator uses an unhalved double sum; the standard symmetric-Laplacian energy includes a 1/2 factor.

4.6 P06: Manifold Regularization

Belkin et al. [2006] develop a learning framework with an ambient regularizer and an intrinsic regularizer. The graph Laplacian term is the finite-sample proxy for intrinsic manifold regularity. In the graph form, large weights penalize variation between neighboring examples, so the conductance/affinity choice controls which functions are preferred.

For SIMODS, P06 supports a regularization interpretation of graph smoothing. The same operator that defines low-pass filtering can also be viewed as a penalty in a learning objective. This is useful for explaining why GCV over graph smoothers is a non-oracle graph-selection candidate: a good graph should make relevant signals predictable or low energy without relying on oracle embedding metrics.

The audit emphasized precision around factor conventions and experimental weight details. The paper’s Eq. (4) prints an unhalved double sum as $f^\top L f$; this should be normalized carefully in synthesis. The WebKB description says weights are based on cosine distances, but the exact affinity transform is not specified in the PDF. Therefore it should not be turned into a precise implementation recipe.

4.7 P07: Graph Signal Processing

Shuman et al. [2013] is the core source for graph-spectral smoothing semantics. A graph Fourier basis comes from Laplacian eigenvectors:

$$L\chi_\ell = \lambda_\ell \chi_\ell.$$

The graph Fourier transform expands a signal in this basis, and a graph filter multiplies each coefficient by a response $h(\lambda_\ell)$:

$$\hat{f}_{\text{filtered}} = \sum_{\ell} h(\lambda_\ell) \hat{f}(\lambda_\ell) \chi_\ell.$$

This is exactly the language SIMODS needs when describing low-pass graph smoothing. The graph, weights, Laplacian normalization, and $h(\lambda)$ together define the smoother.

P07 should be cited primarily for graph-spectral smoothing, not for a particular graph-construction recipe. It also gives a valuable warning: individual eigenvectors inside repeated or nearly repeated eigenspaces are not unique; the meaningful object is the eigenspace or filter action. Comparator discussions should always name the actual spectral response $h(\lambda)$, such as heat, Tikhonov, polynomial, or idealized low-pass.

4.8 P08: Graph Laplacian Convergence on Random Neighborhood Graphs

Hein et al. [2007] analyze convergence of graph Laplacians under random geometric graph sampling, including unnormalized, random-walk, and normalized operators. The paper is a key warning against treating “the graph Laplacian” as a single object. Different normalizations converge to different continuum operators and can place eigenfunction sign changes in different density regions.

For conductance design, P08 matters because it separates graph support, kernel weights, degree normalization, density effects, and limiting operator. If a SIMODS comparator uses a heat kernel, inverse length, or self-tuned kernel, the normalization choice remains a scientific choice. A density-aware operator may be useful for cluster assumptions; a density-invariant operator may be better for geometric recovery.

The audit resolved the mapping to `gflow`: the current `fit.rdgraph.regression()` operator is a custom mass-symmetrized spectral smoother using overlap-density/Riemannian-complex conductance. It is not directly P08’s random-walk, unnormalized, or symmetric-normalized Laplacian.

The audit also notes that $k(0) = 0$ removes self-loops and compact support is part of the main non-compact consistency setup.

4.9 P09: Theoretical Foundation for Laplacian-Based Manifold Methods

Belkin and Niyogi [2008] provide theoretical support for the idea that graph Laplacians built from local kernels can approximate manifold differential operators. The paper is a baseline for heat-kernel and Laplace-Beltrami interpretations of graph smoothness. Its role in SIMODS should be theoretical framing: it does not guarantee finite-sample performance for arbitrary biological graphs or for the current `gflow` overlap-density operator.

The post-audit notes are important. The final synthesis must state the paper’s Laplacian sign convention before comparing equations to heat smoothing or graph-signal filtering. It must also preserve the distinction between theorem settings involving $\text{vol}(M)$ and density scaling. Coifman-Lafon α -normalization can be mentioned contextually here, but detailed α claims should be sourced to P03/P04.

P09 supports the practical reporting rule that any SIMODS kernel-conductance smoother should document four ingredients: distance, bandwidth rule, degree/density normalization, and target operator or practical surrogate.

4.10 P10: kNN Self-Tuned Kernel Graph Laplacian Convergence

Cheng and Wu [2022] connect self-tuned kernels to graph Laplacian convergence with kNN-derived local scales. This is the most direct theory source for the relationship between nearest-neighbor scale estimation and adaptive kernel-conductance rules. The paper separates Dirichlet-form convergence from pointwise operator convergence, a distinction that matters for smoothing: penalty-based smoothers align more naturally with Dirichlet-form statements, whereas pointwise Laplacian estimation needs stronger assumptions.

For SIMODS, P10 supports a planned comparator of the form

$$c_{ij} = \exp\left(-\frac{\ell_{ij}^2}{\widehat{\rho}_i \widehat{\rho}_j}\right)$$

or the paper’s more precise kNN self-tuned scaling. However, kNN bandwidth estimation and graph support construction are different design choices. A SIMODS implementation must decide whether local scales are estimated from all point-cloud neighbors, graph-neighbor edge lengths, or another structure.

The audit also cautions that special α choices are operator-specific. Do not transfer a default from diffusion maps or variable-bandwidth kernels without restating the exact construction. The MNIST/outlier result should be treated as secondary empirical evidence, not convergence proof.

5 Synthesis by Task

5.1 Embedding and Manifold Learning

P02, P03, P04, P08, P09, and P10 all contribute to manifold-learning interpretations, but they emphasize different levels of the pipeline. P02 introduces the graph construction plus generalized eigenproblem. P09 supplies theoretical foundation for local kernel graph Laplacians. P03 adds Markov diffusion and density normalization. P04 and P10 introduce adaptive local scale and show

that local bandwidth changes the limiting operator. P08 warns that normalization and density are not cosmetic.

For SIMODS, this means a graph-smoothing benchmark should not report only “heat kernel” or “self-tuned kernel.” It should report graph support, kernel formula, bandwidth/local-scale rule, degree normalization, and filter response.

5.2 Spectral Clustering

P01 uses local-scale Gaussian affinities to reduce global bandwidth fragility in spectral clustering. P08 gives theoretical background on how graph normalization affects clustering-related eigenfunctions. P02 links Laplacian eigenmaps to spectral clustering through the same graph smoothness objective. These papers support self-tuned and density-aware comparator families, but they do not directly provide a GCV graph-selection criterion.

5.3 Diffusion Geometry

P03 and P04 are the main diffusion-geometry sources. They make clear that kernel weights are only the first stage. After degree and density normalization, the operator may be a Markov transition matrix, an infinitesimal generator, or a symmetric conjugate. Diffusion powers P^t and heat semigroups e^{-tL} are low-pass smoothers with different normalizations.

5.4 Semi-Supervised Learning and Graph Regression

P05 and P06 support the learning/regression view. P05 solves harmonic interpolation on a weighted graph. P06 uses graph Laplacian regularization in a broader learning objective. For SIMODS GCV, these papers justify measuring how well a graph supports smooth prediction of feature signals. They also warn that if the feature task is outcome-specific, the selector may become task-aligned rather than geometry-general.

5.5 Graph Signal Processing and Smoothing

P07 is the clearest conceptual home for low-pass smoothing. It says that the same graph signal can be smooth on one graph and rough on another, because smoothness is defined by the graph operator. This is the logic behind non-oracle graph selection by feature-signal GCV: candidate graphs are scored by whether meaningful signals are low-frequency and predictable. P07 also forces reporting of $h(\lambda)$, not just the graph.

6 Synthesis by Conductance Family

Table 2: Conductance/kernel families and synthesis role.

Family	Formula	Main sources	SIMODS interpretation
Inverse length	$(\ell + \epsilon)^{-\alpha}$	contextual comparator; not a reviewed paper’s central formula	Best direct metric-geometry smoother comparator.

Table 2: Conductance/kernel families and synthesis role.

Family	Formula	Main sources	SIMODS interpretation
Heat/RBF	$\exp(-\ell^2/t),$ $\exp(-\ell^2/(4t))$	P02, P03, P09	Natural for point-cloud diffusion and Laplace-Beltrami approximation.
Exponential/Laplace	$\exp(-\ell/\sigma)$	H005 requested family; less central in Tier 1 corpus	Useful robustness comparator, but needs additional source support.
Self-tuned Gaussian	$\exp[-\ell_{ij}^2/(\sigma_i\sigma_j)]$	P01, P10	Local-scale comparator for nonuniform sampling.
Variable bandwidth	$h(\ell^2/[\epsilon\rho_i\rho_j])$	P04, P10	Theory-backed adaptive bandwidth; operator changes with ρ and normalization.
Density-normalized	$k/(q_i^\alpha q_j^\alpha)$	P03, P04, P08	Separates geometry from sampling density when desired.
Binary support	$c_{ij} = 1$ on graph edges	P02, P08	Control comparator isolating graph support from metric weights.

6.1 Why Not Always Use Inverse Length?

Inverse length is attractive because it respects a direct resistance intuition: short edges should transmit signal strongly; long edges should transmit weakly. For SIMODS, this is exactly why it should be a planned comparator. But the reviewed literature shows why researchers often use other forms:

- Heat kernels approximate local diffusion and suppress long edges exponentially rather than algebraically.
- Gaussian bandwidths provide a scale parameter that defines locality.
- Self-tuned kernels adjust that scale under nonuniform sampling.
- Density normalization can remove sampling-density bias from the limiting operator.
- Markov normalization turns affinities into transition probabilities and makes diffusion distances path-based rather than edge-local.
- Graph signal filters separate graph construction from the spectral response applied to the signal.

Thus inverse length is one scientifically meaningful choice, not the universal default.

7 Theory Results and Their Benchmark Implications

Table 3: Theory themes relevant to SIMODS comparator design.

Source	Setting	Operator issue	Implication
P02	Local graph on point samples	Heat weights and generalized eigenproblem	Heat-kernel weights are principled but bandwidth and graph support remain choices.
P03	Diffusion maps	α -normalization and Markov powers	Degree/density normalization changes the geometry; P^t is a low-pass diffusion smoother.
P04	Variable bandwidth diffusion	$\rho = q^\beta$, α , dimension d	Local scale can stabilize sparse regions but changes drift terms unless normalized deliberately.
P08	Random neighborhood graph convergence	Unnormalized/random-walk/normalized limits	Normalization can move eigenfunctions and density effects; do not claim one generic Laplacian target.
P09	Laplacian-based manifold methods	Kernel graph Laplacian to manifold operator	Kernel-distance smoothers need documented distance, bandwidth, normalization, and target operator.
P10	kNN self-tuned kernels	Local kNN scale and convergence	kNN bandwidth estimation is a serious theory-backed local-scale comparator, especially for Dirichlet-form smoothing.

8 Implications for gflow and SIMODS

8.1 Recommended Comparator Families

The future non-oracle graph-selection benchmark should compare at least:

$$c_e^\rho = \frac{1}{\max(\rho_1(e), 10^{-10})} \quad \text{current `fit.rdgraph.regression()` semantics,} \quad (6)$$

$$c_e^{\text{len}} = \frac{1}{\ell_e + \epsilon} \quad \text{direct metric-length conductance,} \quad (7)$$

$$c_e^{\text{rbf}} = \exp(-\ell_e^2/\sigma^2) \quad \text{heat/RBF conductance,} \quad (8)$$

$$c_e^{\text{exp}} = \exp(-\ell_e/\sigma) \quad \text{exponential-distance conductance,} \quad (9)$$

$$c_{ij}^{\text{self}} = \exp\left(-\frac{\ell_{ij}^2}{\sigma_i \sigma_j}\right) \quad \text{self-tuned local-scale conductance.} \quad (10)$$

For diffusion-style variants, the benchmark must state whether the smoother is formed from a symmetric Laplacian, a random-walk matrix P , a symmetric Markov conjugate, or another mass-symmetrized operator.

8.2 Recommended Diagnostics

The benchmark should not rely on GCV alone. It should record:

- selected graph parameter and selected smoothing parameter;
- GCV score and effective degrees of freedom;
- transfer function $h(\lambda)$;
- eigenvalue spectrum and spectral gaps;
- eigenfunction localization or sign-change diagnostics;
- clean-function RMSE on synthetic known signals;
- agreement with oracle geodesic-isometry performance on synthetic quad benchmarks;
- sensitivity to sampling density and noise.

8.3 How to Phrase Claims in the Manuscript

The manuscript can safely say:

- Graph smoothing is operator-dependent: the graph support, edge weights, normalization, and spectral response define what “low frequency” means.
- Heat/RBF, self-tuned, and density-normalized kernels are standard families for graph Laplacians and diffusion operators.
- Local-scale kernels are motivated by nonuniform sampling and can change both numerical behavior and continuum limits.
- The current `gflow` smoother is an overlap-density / Riemannian-complex smoother, not a direct length-kernel smoother.
- A controlled SIMODS benchmark should test whether current overlap-density smoothing and length/kernel-conductance smoothing select graph parameters that agree with oracle geodesic-isometry performance.

The manuscript should avoid saying:

- that `fit.rdgraph.regression()` currently implements $1/(\ell + \epsilon)$, Gaussian, or self-tuned conductances;
- that self-tuned bandwidth estimation is the same as cKNN graph support;
- that one α or Laplacian normalization has the same meaning across all reviewed papers;
- that asymptotic manifold theory guarantees finite-sample SIMODS performance without a matching benchmark.

9 Evidence Label Table

Table 4: Selected synthesis claims with evidence labels.

Claim	Label	Source	Notes
Self-tuned spectral clustering uses $\exp[-d^2/(\sigma_i\sigma_j)]$.	explicit	P01, Eq. (1)	Local scale from K th neighbor distance.
Laplacian Eigenmaps use heat weights and $Ly = \lambda Dy$.	explicit	P02	Heat bandwidth conventions t and $4t$ both appear.
Diffusion maps define diffusion distance through P^t spectra.	explicit	P03	Markov normalization central.
Increasing diffusion time damps smaller Markov eigenvalues.	derived	P03 Figure 3 and formulas	Low-pass language is synthesis.
Variable bandwidth kernels change continuum drift terms.	explicit	P04	Depends on α, β, d .
Harmonic graph interpolation depends on edge weights.	derived	P05	Weight-as-conductance language safest in electrical interpretation.
Graph filters require reporting $h(\lambda)$.	derived/contextual	P07	Essential for SIMODS comparator reporting.
Normalization changes graph Laplacian limits and eigenfunctions.	explicit/derived	P08	Avoid generic “the graph Laplacian.”
Kernel graph Laplacian theory supports planned heat-kernel comparators.	derived	P09	Applicability depends on assumptions.
kNN self-tuned kernels support adaptive local-scale comparators.	derived	P10	Distinguish bandwidth estimation from support construction.
Current gflow smoother is overlap-density, not length/RBF/self-tuned.	contextual	H003/H004 source audit plus H005 scope	Not a claim from the reviewed papers.

10 Open Risks and Questions

1. The Tier 1 corpus is strong for heat/RBF, diffusion, self-tuned, local bandwidth, and graph signal filtering. It is weaker for direct inverse-length and $\exp(-\ell/\sigma)$ conductance rules as primary literature topics. Those may need Tier 2 additions if the final manuscript foregrounds them.
2. Several sources use different Laplacian sign conventions. Final equations should state whether L is positive semidefinite and whether heat is e^{-tL} or $e^{t\Delta}$ under a PDE sign convention.

3. Local bandwidth and graph support are separate implementation choices. A SIMODS self-tuned comparator should define both.
4. If SVG explanatory figures are used in a `pdflatex` report, they should be rendered to PDF or PNG.
5. Reproduced paper-figure cropped figure panels are now stored under `paper_figures/` and tracked by `paper_figure_screenshots.yml`. These are internal-review screenshots from canonical reading copies; if the final public supplement uses paper figures, permissions and reproduction labels will still be needed.

11 Conclusion

The literature supports a principled benchmark in which the current overlap-density/Riemannian-complex `gflow` smoother is compared against direct length and kernel conductance smoothers. The most important design lesson is that the conductance rule cannot be separated from graph support, normalization, spectral response, and task. Heat kernels emphasize local diffusion; self-tuned kernels adapt to local sampling scale; variable-bandwidth and density-normalized kernels change continuum limits; graph signal processing turns the resulting operator into a filter $h(\lambda)$. A SIMODS non-oracle graph-selection criterion should therefore report the full operator construction and should judge candidate smoothers by agreement with oracle geodesic-isometry performance, signal reconstruction behavior, and robustness diagnostics rather than by GCV alone.

A Local Artifact Inventory

The H005 workspace contains:

- `source_manifest.yml`: source acquisition manifest.
- `references.bib`: BibTeX database for this report.
- `paper_memos/P*.md`: revised-after-audit paper review memos.
- `audits/P*_audit.md`: paper-specific audit memos.
- `sources/pdf/P*.pdf`: cached canonical Tier 1 reading copies.
- `figures/P01_self_tuned_kernel_toy.png`, `figures/P04_local_bandwidth_conductance_curve.png`, `figures/P05_three_node_harmonic_interpolation.svg`, `figures/P07_graph_filter_pipeline.png`, and `figures/P09_operator_limit_map.svg`: original internal explanatory figures created by reviewers.

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